

$$\pi_j^{(n)} := P(X_n = j)$$

$$n \rightarrow \infty$$

Ejemplo donde no hay distrib. límite

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

finito ✓
irreducible ✓
⇒ rec. positivo ✓

◦◦ La dist. estacionaria es
única y está dada por

$$\pi_j = \frac{1}{m_j}$$

3 resultados límite

Dist. límite

$$\lim_{n \rightarrow \infty} \pi^{(n)}$$

Dist. invariante

$$\pi P = \pi$$

Teo Ergódico

$$\pi = \left(\frac{1}{m_0}, \dots, \frac{1}{m_m} \right)$$

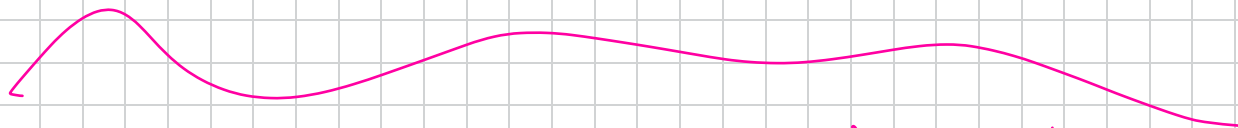
Prop 1.17

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$$\lim_{m \rightarrow \infty} P_{kj}^{(m+1)} = \sum_{i=0}^N \lim_{m \rightarrow \infty} P_{ki}^{(m)} P_{ij}$$

$$\pi_j = \sum_{i=0}^N \pi_i P_{ij}$$

$$\pi = \pi P$$



$$\sum_{j \in S} \pi_j = \sum_{j \in S} \lim_{n \rightarrow \infty} P_{ij}^{(n)} \quad \text{p.a. } i$$
$$= 1$$

Prop 1.17 (Cadena con infinitos estados)

$$\pi_j \geq \sum_{k \in S} \pi_k p_{kj}$$

sup $\exists j_0$ para el cual

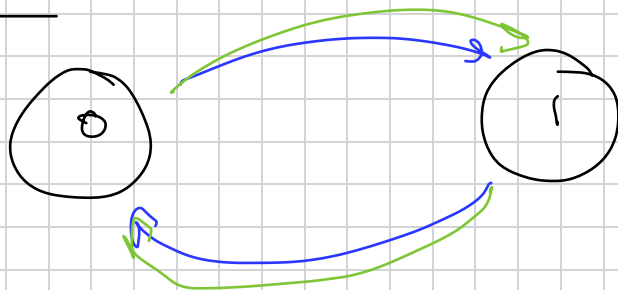
$$\pi_{j_0} > \sum_{k \in S} \pi_k p_{kj_0}$$

$$\Rightarrow \sum_{j \in S} \pi_j > \sum_{j \in S} \sum_{k \in S} \pi_k p_{kj}$$

$$\sum_{j \in S} \pi_j > \sum_{k \in S} \pi_k \quad \nabla$$

$$\circ \circ \quad \forall j \quad \pi_j = \sum_{k \in S} \pi_k p_{kj}$$

Cadena periódica $d=2$



$$\underline{d(0) = d(1) = 2}$$

$$p_{00}^{(1)} = 0$$

$$p_{00}^{(2)} = 1$$

$$p_{00}^{(3)} = 0$$

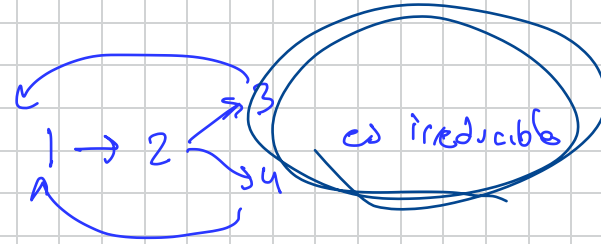
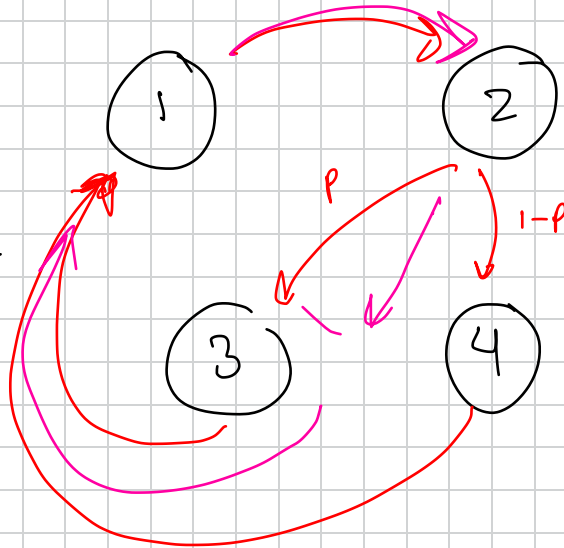
$$p_{00}^{(4)} = 1$$

Ordena periódica $d=3$

• Finitas

• Rec. Positivas

• $\exists!$ π distrib.
estacionaria



$$d(1) = 3$$

mod $\{3, 6, 9, 12, \dots\}$